

Homework 5

Solutions must be **typed** and **submitted electronically**.

1. Let $G : \{0, 1\}^\lambda \rightarrow \{0, 1\}^{3\lambda}$ be a secure length-**tripling** PRG. For each function below, state whether it is also a secure PRG. If the function is a secure PRG, give a proof. If not, then describe a successful distinguisher and explicitly compute its advantage.

(a)
$$\begin{array}{l} \text{H(s):} \\ x := G(s) \\ y := G(0^\lambda) \\ \text{return } x \| y \end{array}$$

(b)
$$\begin{array}{l} \text{H(s):} \\ x := G(s) \\ y := G(0^\lambda) \\ \text{return } x \oplus y \end{array}$$

(c)
$$\begin{array}{l} \text{H(s):} \\ x \| y \| z := G(s) \\ w := G(x) \\ \text{return } x \| y \| z \| w \end{array}$$

(each of x, y, z are λ bits long)